Small Models that Forget: Exploiting Temporal Locality with Online Learning at Inference Time

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Abstract

Many inputs that machine learning models see at inference time have strong temporal locality: consecutive inputs tend to be similar and come from a small part of the input distribution. A model facing such inputs does not need to keep everything it knows ready at all times. It may be enough to remember some basic knowledge and adapt quickly to the current context. We study whether this lets small models beat their frozen offline version. Small multilayer perceptrons, trained offline on MNIST digits under eight rotations, classify a stream of test images arranged in bursts: every burst holds all ten digits under one rotation, so the locality is in the inputs and the recent labels carry no information. The models predict each image and then take one update on it. A shuffled copy of the same stream serves as a control. Plain online SGD barely beats the frozen model (at most +0.25 percentage points on bursts of 20 to 200 images, +1.6 on bursts of 1000). Slow and fast weights gain up to +6.1 points on bursts of 1000 but lose on short bursts, where each context switch costs more than the adaptation recovers. Meta-learning the slow weights together with per-parameter fast learning rates and decays (Meta-SGD) is the only method that also gains on short bursts (+2.4 to +2.7 points), still far from the +19 to +23 points that a model per rotation would give.

1 Introduction

A trained network is usually deployed frozen: it must classify any input from the whole distribution it was trained on. In many applications, however, the inputs arrive in a stream with strong temporal locality. A camera in a kitchen sees kitchen objects for hours; a speech recognizer hears the same speaker for a whole conversation; a sensor stays in the same regime for long periods. At any moment only a small part of the input distribution is active.

Computer memory hierarchies have relied on this property for decades: a small cache that holds the recently used data serves most requests [2]. We ask whether the same idea applies to the knowledge stored in a network. A model does not need to remember everything all the time if its inputs have strong temporal locality. It may only need some basic knowledge plus the ability to adapt fast to the current context.

Our hypothesis is that, under severe capacity limits, online training at inference time can outperform the same model trained offline and then frozen. A frozen small model must spread its limited capacity over the whole distribution. An online model can spend it on the part of the distribution that is active now. The price is forgetting: neural networks trained sequentially lose what they learned earlier, a behavior known as catastrophic forgetting [13, 4]. This recency bias is usually treated as a problem. In a stream with temporal locality it may be useful, because what was seen recently is what will be seen next.

This paper tests the hypothesis on MNIST [10]. The obvious stream, bursts of plain MNIST images of a few digits each, puts the locality in the labels only: the images of a burst share nothing but their digits. On such a stream a running estimate of the recent label frequencies, which learns no weights, beats online learning, so the stream says little about whether a network can adapt its knowledge. We therefore put the locality in the inputs: every burst holds all ten digits under one of eight rotations, and the labels are uniform everywhere. To gain, a model has to adapt to how the current inputs look, not to which labels are frequent.

Experiment 1 checks whether plain online SGD beats the frozen model on two streams, with short bursts (20 to 200 images) and long ones (1000). Experiment 2 splits knowledge by time scale, with slow weights and fast weights that adapt to the current context and then fade. Experiment 3 meta-learns the slow weights together with per-parameter fast learning rates and decays (Meta-SGD [11]).

2 Related work

Catastrophic forgetting and continual learning. Continual learning tries to learn a sequence of tasks without forgetting the earlier ones [14]. Elastic weight consolidation [8], for example, protects the weights that matter for old tasks. Our setting reverses the goal: we accept forgetting and ask whether it can be used.

Two time scales. Hinton and Plaut [6] added to each connection a fast weight that learns quickly and decays, on top of a slow weight, to recover old memories after interference. Complementary learning systems theory [12] explains the division between a fast-learning hippocampus and a slow-learning neocortex by the need to learn new experiences without overwriting old knowledge. Fast weights in recurrent networks [1] store recent hidden states in a decaying outer-product memory. Experiment 2 uses the formulation of Hinton and Plaut on the first layer only, and its fast part tracks the current context rather than restoring old associations.

Meta-learning for fast adaptation. Model-agnostic meta-learning [3] learns an initialization from which a few gradient steps adapt a network to a new task. Meta-SGD [11] also learns a learning rate for every parameter. Experiment 3 meta-learns, in the same way, the slow weights and a per-parameter fast learning rate and decay, so that one gradient step per image adapts the model to the current context.

Test time adaptation. Several methods update a model while it is used, for example with a self supervised task [15] or by minimizing the entropy of the predictions [16]. They target distribution shift and do not use labels. Dynamic evaluation [9] updates a language model on the recent part of the sequence it is predicting, which exploits local regularities in text, the closest idea to ours.

Model compression. Compression reduces model size by pruning, quantization or distillation [5], keeping the model frozen. We study a different route to small models that is complementary to compression.

3 Method

3.1 Contexts and streams

A context is one of 8 rotations of the image, by $k \cdot 22.5$ degrees for $k = 0, \ldots, 7$ (bilinear, black outside the image), applied before normalization. A stream is a sequence of MNIST test images, each shown under the context of its burst. Every burst holds all ten digits under one context, and consecutive bursts have different contexts. The labels are uniform everywhere, so the recent labels carry no information.

The *short* stream has 10,000 positions, with every test image once in a random order, split into 88 bursts of 20 to 200 images (113.6 on average); each context starts 4 to 14 bursts. The *L1000* stream has 80,000 positions, with every test image once under every context, in 80 bursts of exactly 1000 images, 10 per context. In L1000 every test image appears 8 times, once per rotation.

Each stream has a control: a random permutation of the same positions, with each image keeping its context. It contains exactly the same images, so any difference in accuracy comes from their order. The control is not a scenario we expect in practice: we assume that the real input streams have temporal locality and are not random. Its only purpose is to show whether a gain from online learning comes from that locality or merely from training on more data, including, in L1000, learning the repeated images.

3.2 Protocol

Each model is first trained offline and then processes a stream one image at a time. For each image, the model first predicts its label, and the prediction counts towards the online accuracy. Then it takes one update on that single image with its true label. This assumes that the label becomes available after each prediction, as in prequential evaluation. Each stream starts from the same offline weights.

We report these quantities, all on the same positions of a stream:

- *frozen accuracy*: the offline model without updates (equal on a stream and its control, since they hold the same images under the same contexts);
- *online accuracy on bursts* and *on the shuffled control*;
- *online gain*: online accuracy on bursts minus frozen accuracy;
- *locality effect*: online accuracy on bursts minus online accuracy on the shuffled control.

Differences are given in percentage points (pp). The online gain tells whether online learning is worth it. The locality effect isolates the part of the gain that comes from temporal locality, removing the effect of simply training more and of seeing images more than once.

3.3 Model and baseline

The model is a multilayer perceptron with one hidden layer of h ReLU units, $784 \to h \to 10$, with $h = 8$ (24.9 KB as 32 bit floats) and $h = 4$ (12.5 KB). The baseline is the *mixed* model: the same network trained offline on the 60,000 training images, each sampled image shown under a context drawn at random, with Adam [7] (learning rate 0.001, batch 128, 50 epochs, seed 0). On the test set, averaged over the contexts, it reaches 74.05% ($h = 8$) and 64.78% ($h = 4$). A model of the same size trained on one context only reaches 93.26% and 87.31% on it, so there is roughly +19 to +23 pp to gain by adapting to the current context.

3.4 Online methods

All three methods share one update. The network predicts with $S + F$, where S starts at the mixed model and F at zero. After each image, with g the gradient of its loss at $S + F$,

$$S \leftarrow S - \eta_{\text{slow}}\, g,$$
$$F \leftarrow \lambda F - \eta_{\text{fast}}\, g,$$

the second line only on the parameters that have a fast part. The fast part is a decaying sum of recent gradients: it follows the current burst and forgets it within about $1/(1-\lambda)$ images.

- *SGD* (Experiment 1): plain SGD, no fast part.
- *Slow and fast weights* (Experiment 2): a fast part on the first layer, weights and bias.
- *Meta-SGD* (Experiment 3): $\eta_{\text{slow}} = 0$ and a fast part on every parameter, with η_{fast} and λ replaced by per-parameter tensors $\alpha > 0$ and $0 < \lambda < 1$.

For Meta-SGD, S, α and λ are meta-learned [11] on training images only, on episodes of 3 bursts of 20 to 60 images under different contexts. The meta-loss is the mean prediction loss over the episode, backpropagated through the whole online unroll; the per-image gradient is written in closed form, so the meta-gradient is exact. Meta-training uses Adam (learning rate 0.001), 32 episodes per step, gradient clipping at norm 1, $\alpha = 10^{-3}$ and $\lambda = 0.99$ at the start, S starting at the mixed model, and early stopping on 256 held-out episodes, with at most 15 minutes per model; every run reached that limit (4800 to 5850 steps). A second variant keeps S fixed at the mixed model and learns only α and λ. The same checkpoints are evaluated on both streams.

Each method is run at a handful of settings, compared on the evaluation streams themselves. All runs use seed 0.

4 Experiments

Table 1 gives the results of the three experiments, Figure 1 compares the best setting of each method and Figure 2 shows how accuracy evolves within a burst.

4.1 Experiment 1: online SGD

On the short stream, plain online SGD stays within ±0.25 pp of the frozen model at small learning rates and loses 2 to 7 pp at larger ones. On L1000 it gains at most +0.69 pp ($h = 8$) and +1.58 pp ($h = 4$). Larger learning rates have a larger locality effect (up to +6.66 pp): they adapt within a burst and reach 75.6% ($h = 8$) and 68.4% ($h = 4$) after the 200th image at $\eta = 3 \cdot 10^{-3}$. But they lose most of it after each context switch, falling to 50–58% over the first 10 images of an L1000 burst. With small learning rates, part of the L1000 gain is not locality: the shuffled control also beats the frozen model, by learning the repeated images.

4.2 Experiment 2: slow and fast weights

A fast first layer gains clearly on L1000: +4.11 pp ($h = 8$) and +6.11 pp ($h = 4$) with $\eta_{\text{fast}} = 10^{-2}$ and $\lambda = 0.999$, with a locality effect of +8.47 and +10.22 pp. On the short stream it loses 3 to 4 pp: accuracy falls to 56–67% over the first 50 images of a burst and needs about 100 images to recover, longer than most short bursts. Settings with a smaller fast learning rate lose less after a switch but

Table 1: Online gain (bursts minus frozen) / locality effect (bursts minus shuffled), in pp. Short: bursts of 20 to 200 images; L1000: bursts of 1000. Experiment 2: η_{slow}, η_{fast}, λ, fast part on the first layer.

		short		L1000	
	setting	$h = 8$	$h = 4$	$h = 8$	$h = 4$
	frozen mixed model (%)	73.25	64.41	74.05	64.78
1	SGD, $\eta = 10^{-5}$	+0.25 / +0.21	+0.09 / +0.05	+0.53 / +0.10	+0.37 / −0.01
	SGD, $\eta = 10^{-4}$	+0.17 / +0.00	−0.15 / −0.01	+0.69 / +0.35	+0.66 / +0.33
	SGD, $\eta = 10^{-3}$	−2.88 / +0.61	−2.40 / +0.39	+0.20 / +2.81	+1.58 / +3.24
	SGD, $\eta = 3 \cdot 10^{-3}$	−7.03 / +0.86	−5.13 / +1.28	−0.63 / +5.89	+1.41 / +6.66
2	10^{-4}, 10^{-3}, 0.99	+0.28 / +0.29	+0.09 / +0.26	+1.10 / +0.93	+1.43 / +1.33
	10^{-4}, $3 \cdot 10^{-3}$, 0.997	+0.07 / +1.12	−0.36 / +0.99	+2.92 / +3.64	+3.99 / +4.90
	10^{-4}, 10^{-2}, 0.999	−3.47 / +2.09	−3.07 / +2.47	+4.11 / +8.47	+6.11 / +10.22
	0, 10^{-2}, 0.999	−3.66 / +2.04	−3.77 / +3.06	+2.87 / +8.81	+3.02 / +9.57
3	Meta-SGD	+2.70 / +3.56	+2.36 / +4.05	+3.51 / +3.99	+3.77 / +4.41
	Meta-SGD, S fixed	+1.19 / +1.20	+1.07 / +1.31	+1.89 / +1.99	+1.87 / +2.05

gain less on long bursts; none gains more than +0.3 pp on the short stream. With the slow learning rate at 0 the gain on L1000 drops to +2.9 and +3.0 pp: the slow part contributes +1.2 to +3.1 pp, partly by learning the repeated images.

4.3 Experiment 3: Meta-SGD

Meta-SGD is the only method that gains on the short stream: +2.70 pp ($h = 8$) and +2.36 pp ($h = 4$), with locality effects of +3.56 and +4.05 pp. Its switching cost is small: over the first 10 images of a short burst it is 0.6 to 1.2 pp below the frozen model, and above it from the 11th image on. On L1000 it gains +3.51 and +3.77 pp, less than the best fast first layer, because its accuracy stops rising after about 50 images: the meta-learned decays (median λ between 0.74 and 0.97 depending on the layer) forget faster than a burst of 1000 needs, since meta-training only saw bursts of 20 to 60. Learning S matters: with S fixed the gain roughly halves (+1.07 to +1.89 pp). Yet the meta-learned S alone, frozen, is within −0.3 to +0.5 pp of the mixed model, so learning S mostly makes the model easier to adapt rather than more accurate by itself.

5 Discussion

With the locality in the inputs, plain online SGD is not worth it: the frozen mixed model is hard to beat, even though a model per rotation would be 19 to 23 pp more accurate. What decides the outcome is the cost of each context switch against the length of the context. Seen this way, the fast learning rate and the decay set the length of the model's memory: a faster setting adapts faster and forgets faster. A fast first layer that adapts strongly pays off on bursts of 1000 images, but after a switch it is worse than the frozen model for about 100 images, so it loses on bursts of 20 to 200. Meta-learning the fast learning rates, the decays and the slow weights produces a model that adapts within a few images and loses little at a switch, and it is the first method to gain on short bursts. But it captures only a small part of the headroom, and it is tied to the burst lengths it was meta-trained on.

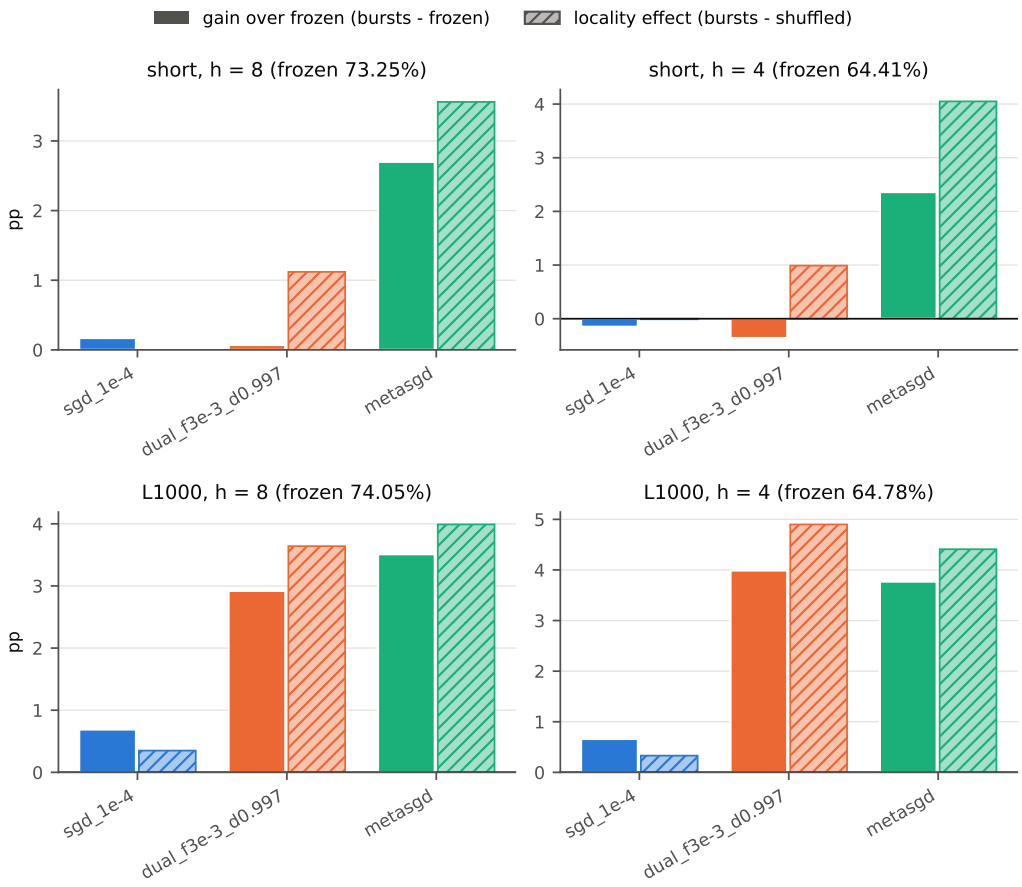


Figure 1: Online gain (solid) and locality effect (hatched) of the best setting of each method, chosen by mean gain over sizes and streams. Blue: SGD ($\eta = 10^{-4}$); orange: slow and fast weights (10^{-4}, $3 \cdot 10^{-3}$, 0.997); green: Meta-SGD.

The shuffled control is not a deployment scenario: we assume that real streams have temporal locality. It shows that every way of exploiting the bursts more strongly, from a larger learning rate to fast weights, relies more on that locality and would not help on a stream without it.

The results also have clear limits. The task is MNIST and the stream is synthetic, with an idealized form of locality: bursts with sharp boundaries and a single rotation each. The online model receives the true label right after each prediction, which many applications cannot provide. Every result comes from a single run with seed 0, so differences below about 0.5 pp may not be meaningful. The settings were compared on the evaluation streams, so the best ones are slightly optimistic, and meta-training was limited to 15 minutes per model and had not converged. The L1000 stream shows every test image 8 times; the shuffled control has the same repetitions, so the locality effect cancels them, but the online gain does not.

6 Conclusion and next steps

When the locality of a stream is in the inputs, small models gain little from plain online learning. Plain SGD barely beats the frozen model trained on all rotations, a fast first layer gains up to +6.1 pp only when the bursts are long (1000 images), and Meta-SGD is the only method that also

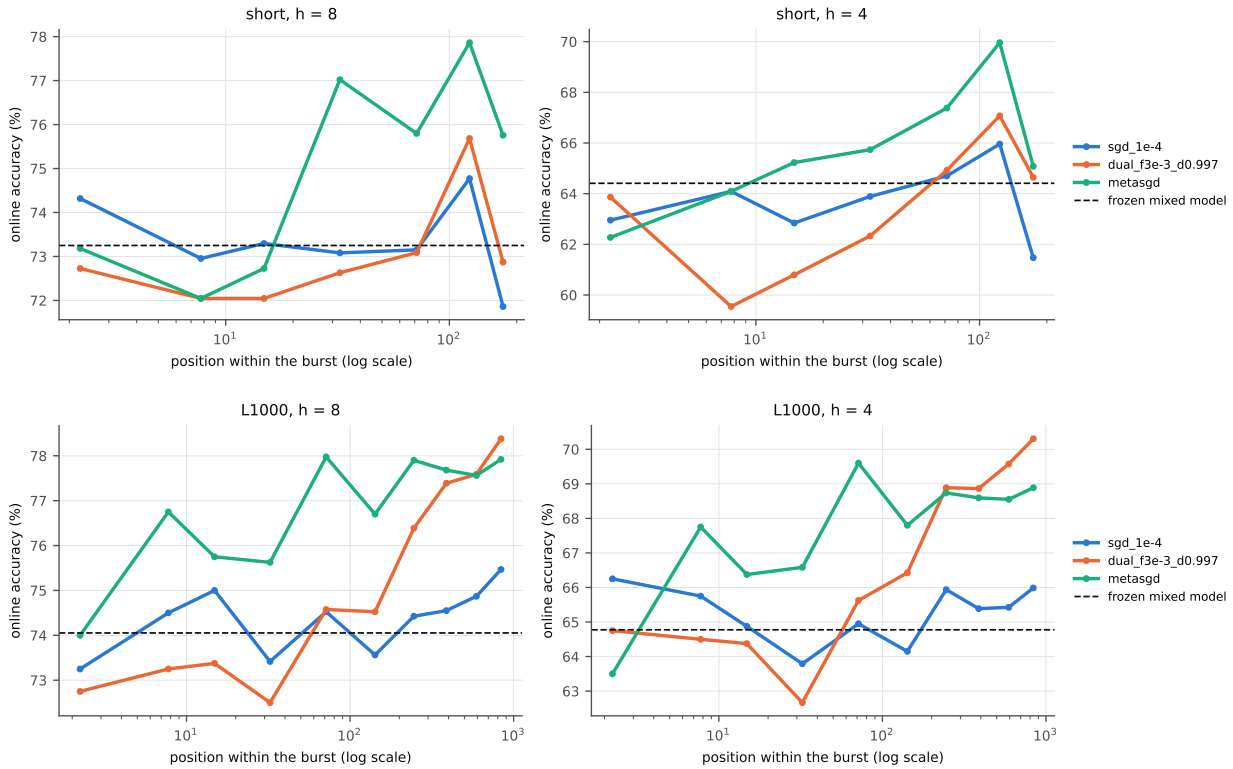


Figure 2: Online accuracy against position within the burst (binned, log scale) for the settings of Figure 1, on the short stream (top) and L1000 (bottom). Dashed: the frozen mixed model.

gains on bursts of 20 to 200 images (+2.4 to +2.7 pp). The cost of a context switch, not the speed of adaptation alone, is what limits online learning on short contexts.

Next steps include lowering that cost (for example by resetting the fast part when the model detects a switch), meta-training on episodes as long as the bursts or with a floor on the decay, repeating the runs with several seeds, tuning on separate streams, testing streams with natural locality (for example video) and other architectures, and replacing the true labels with weaker feedback.

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